

ADITYA VERMA

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks(%)
2021 - 2025	B.TECH in Computer Science and Engineering	Graphic Era University, Dehradun	8.32

EXPERIENCE

Software Engineer, TCS -Mumbai, MH November, 2025 - Current

- Designed and developed scalable backend systems using Python and PySpark for the Reserve Bank of India's Centralised Information Management System (CIMS) modernization, migrating 200+ regulatory workflows across 5,000+ financial institutions from report-based to element-based data architecture.
- Architected bi-directional data processing frameworks covering ingestion of structured SDMX data and dynamic generation of regulatory reports, powering both real-time query platforms and end-user reporting systems.
- Built high-performance distributed data processing modules with Apache Spark, optimizing data transformation, validation, and mapping logic to ensure reliability, consistency, and regulatory compliance at scale.
- Engineered fault-tolerant data pipelines and backend services integrating SQL and Hive, improving performance and maintainability in large-scale production environments.
- Led the development and debugging of large-scale data migration systems, identifying and resolving system-level defects, ensuring accurate and consistent transformation of legacy datasets into the new architecture.
- Collaborated with cross-functional teams in an Agile environment to deliver business-critical features, participating in the full development lifecycle from design to deployment.

PROJECTS

WordSphere: A comprehensive blogging application (Aug, 2024 - Sept, 2024) : word-sphere.vercel.app

- Developed a complete **full-stack** blogging application using **React.js** (frontend) and **Node.js + Express.js** (backend) with **MongoDB** for data persistence.
- Implemented **REST APIs** for CRUD operations (post creation, editing, deletion), user authentication, and advanced content filtering.
- Integrated **TinyMCE** rich-text editor and modern styling with **Tailwind CSS** for excellent user experience and accessibility.
- Deployed on Vercel, demonstrating end-to-end ownership from architecture to production.

Self-Driving Car Agent (Sept, 2023 - Dec, 2023) : github.com/addy-301/Self-Driving-Car-Agent-with-RL

- Developed an autonomous driving agent in **Unity3D ML-Agents** using **Proximal Policy Optimization (PPO)** and **Soft Actor-Critic (SAC)** algorithms.
- Created simulation environments for training and testing the agent, achieving **92% decision-making accuracy** and fast adaptation (3.5s average) to dynamic conditions.

Real-Time Vehicle Tracking System (Aug, 2023 - Sept, 2023) : github.com/addy-301/RealTime-Vehicle-Tracking-System-SIH2023

- Engineered a real-time vehicle tracking web application achieving 98% real-time update accuracy.
- Created a user-friendly interface for live vehicle location and status monitoring.

ACHIEVEMENTS

- TCS AI Fridays'** - Won 1st place among 15 teams at TCS AI Fridays internal AI hackathon, building a working, well-documented Retrieval-Augmented Generation (RAG) application in 7 hours.
- Graphic Era Coding Competition(ICCC)** - Secured 3rd rank in a coding contest hosted by Graphic Era.
- TechWizard** - Qualified twice for Round 2 of TechWizard, an inter-university coding contest at Graphic Era University.
- INFathon** - Qualified for Round 2 of INFathon, a code-a-thon organized by Informatica.

SKILLS

Programming Languages	Python, Java, JavaScript, HTML, CSS, C++
Web Development	React.js, Node.js, Express.js, REST APIs, Tailwind CSS
Database	MySQL, MongoDB, Apache Hive, Mongoose
AI / Machine Learning	Generative AI, Retrieval-Augmented Generation (RAG), Agentic AI, Reinforcement Learning
Cloud & Tools	Google Cloud Platform (GCP), Git, GitHub, Docker
Core Concepts	SDLC, Agile, Object-Oriented Programming (OOP), Data Structures and Algorithms

CERTIFICATIONS

- **Salesforce Agentblazer Innovator - Salesforce**
- **AWS GenAI Practitioner - Amazon Web Services (AWS)**
- **Introduction to Generative AI - Google Cloud**
- **Cloud Computing Foundations: Cloud Computing Fundamentals - Google Cloud**